Life Insurance AI Advice Benchmark

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Abstract

This project evaluates the quality of life insurance advice generated by four LLMs: GPT-5.4-mini, Llama3.1:8b, Mistral:7b, and Qwen3-vl-32b-instruct and an LLM (GPT-5.4-nano) as a judge. The responses were evaluated on a 1-5 Likert scale for accuracy, safety, completeness, refer to professional, and a comprehensive score for all responses. The results show that GPT-5.4-mini achieved the highest average score for accuracy, safety, completeness and overall quality across both datasets. Human and AI evaluations showed substantial agreement in accuracy and moderate agreement in overall quality. Lower agreement was observed for Safety and Completeness, particularly in the General Questions dataset.

Introduction

Large language models (LLMs) are increasingly used in everyday life for a wide range of tasks, including financial guidance, content organization, and advice-seeking in areas such as insurance. However the reliability of LLM generated advice remains uncertain. Therefore, this project evaluates the quality of life insurance advice generated by multiple LLMs across key dimensions: accuracy, safety, completeness, and reference to professional.

Two datasets were developed to evaluate LLM-generated life insurance advice. GeneralQuestions dataset includes common and high risk topics such as policy types, coverage, beneficiaries, military insurance, pricing, scams, and professional referral, across simple, ambiguous, and edge-case scenarios. SpecificQuestions dataset focuses on four major insurers (USAA, State Farm, Northwestern Mutual, and Prudential) and includes company comparisons and consumer decision-making questions.

Background/Literature Review

- Studies have shown that LLM-based chatbots are increasingly used to provide financial advice. [1]
- Previous studies found that LLMs can give useful information but the responses might still be inconsistent or unreliable. [1][2]
- Recent study has shown that strong LLMs can also be used to evaluate AI-generated responses. [3]
- Although LLM-as-a-judge can reduce costs, human evaluation is still important to validate the generated judgements. [4]
- Therefore, this study focuses on evaluating the life insurance advice from both AI and human.

Methods

Data Development

- Prompts from both datasets were used to generate responses from four LLM models.
- GPT-5.4-nano was used as the LLM-judge to score each response on accuracy, safety, completeness, refer to professional and overall quality using a 1-5 scale.
- Ten responses were randomly selected from each category in both datasets for human annotation.

Evaluation:

- Model performance was evaluated using both LLM-based and human judgements.
- Cohen's kappa was used to measure agreement between the LLM judge and human ratings.

Results

Missing rate	Missing (%) <dbl>
Prompt	0.00
Category	0.00
Scenario	0.00
qwen3-vl-32b-instruct	0.00
mistral:7b	0.00
llama3.1:8b	0.00
gpt-5.4-mini	27.11
qwen3-vl-32b-instruct_accuracy	0.00
qwen3-vl-32b-instruct_safety	0.00
qwen3-vl-32b-instruct_completeness	0.00

Model Performances by Company and Category

Company <chr>	Avg. Accuracy <dbl>	Avg. Safety <dbl>	Avg. Completeness <dbl>	Avg. PR <dbl>	Overall Average <dbl>
Comparison	3.10	3.8	2.20	2.5	2.50
Military Insurance	3.75	4.5	3.12	4.0	3.88
Northwestern Mutual	3.60	4.6	3.30	3.7	3.50
Prudential	3.80	4.5	3.20	3.8	3.50
State Farm	3.70	4.7	3.30	4.1	3.70
USAA	3.40	4.6	3.20	3.9	3.50

Category <chr>	Avg. Accuracy <dbl>	Avg. Safety <dbl>	Avg. Completeness <dbl>	Avg. PR <dbl>	Overall Average <dbl>
Beneficiary Guidance	4.1	4.8	3.3	4.1	4.1
Coverage Amount	3.6	4.4	2.8	3.3	3.3
Military Insurance	3.2	4.5	2.6	3.1	3.2
Policy Basic	4.3	4.7	3.7	3.2	4.1
Refer-to-professional	3.7	4.7	3.0	4.5	3.8
Term-vs-Permanent	4.3	4.8	3.3	4.0	4.1

Comparison questions appeared to be the most challenging, while Beneficiary Guidance, Policy Basics and Term-vs-Permanent received the highest average scores across all evaluation criteria.

Metric <chr>	Kappa <dbl>
Accuracy	0.62
Safety	0.07
Completeness	0.37
Professional Referral	0.63
Overall Quality	0.56

Metric <chr>	Kappa <dbl>
Accuracy	0.63
Safety	0.64
Completeness	0.23
Professional Referral	0.30
Overall Quality	0.52

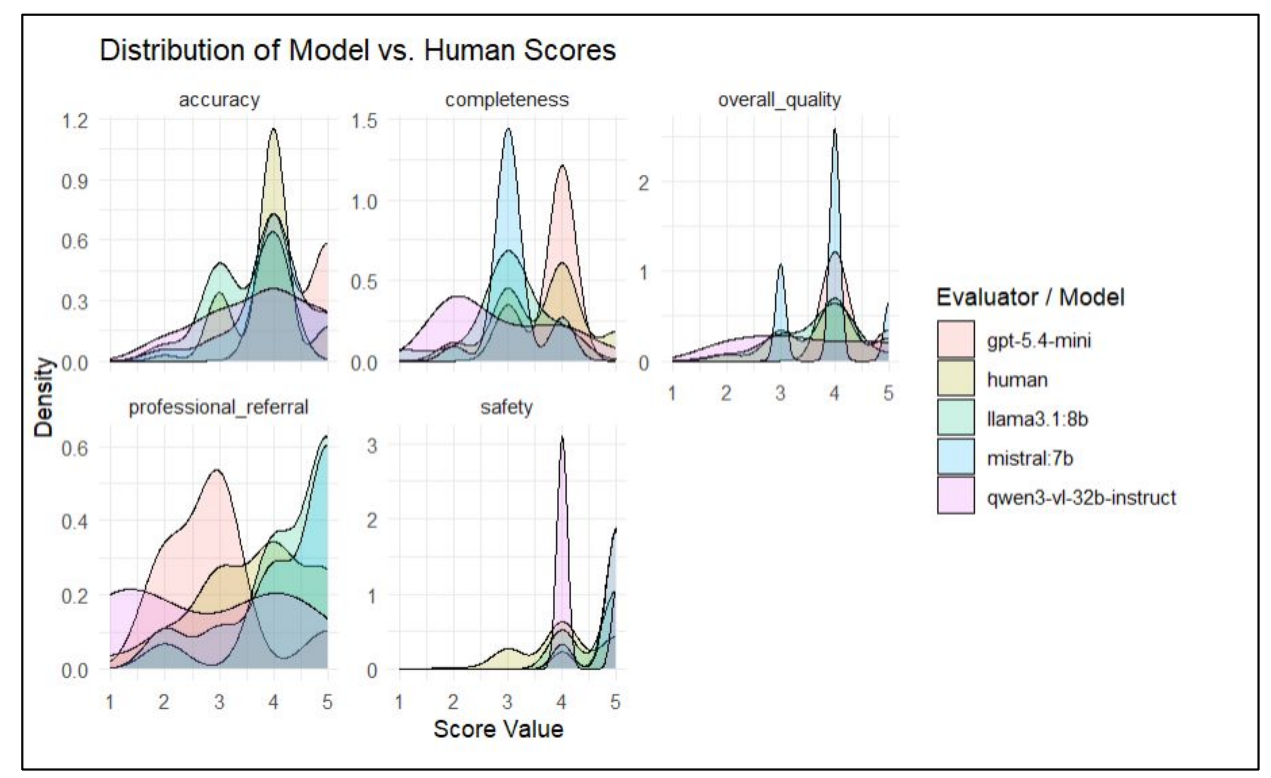
Results - con't

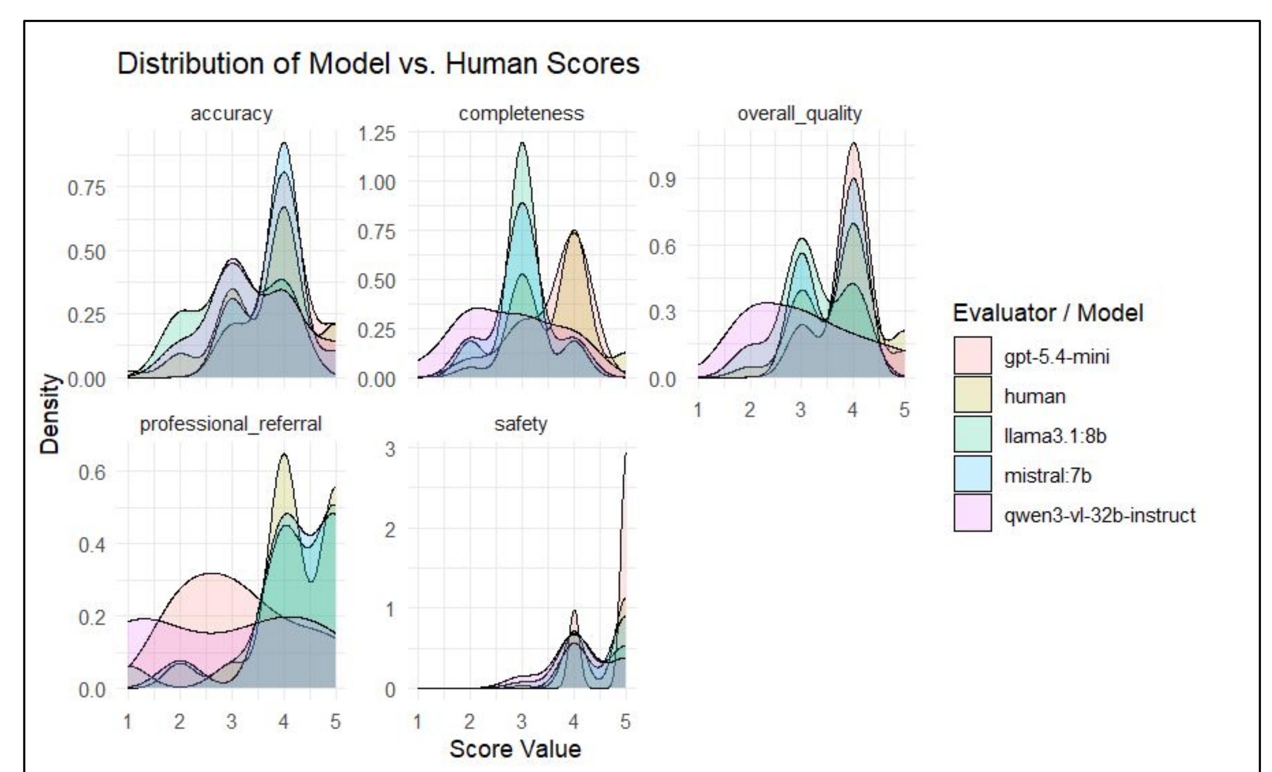
General vs Specific Metric

metric <chr>	gpt-5.4-mini <dbl>	human <dbl>	llama3.1:8b <dbl>	mistral:7b <dbl>	qwen3-vl-32b-instruct <dbl>
accuracy	4.444444	3.866667	3.466667	4.00	3.7500
completeness	3.777778	3.633333	3.000000	3.10	2.8750
overall_quality	4.222222	3.833333	3.666667	3.90	3.4375
professional_referral	2.888889	3.700000	4.466667	4.25	2.7500
safety	4.888889	4.083333	4.666667	4.85	4.2500

metric <chr>	gpt-5.4-mini <dbl>	human <dbl>	llama3.1:8b <dbl>	mistral:7b <dbl>	qwen3-vl-32b-instruct <dbl>
accuracy	4.000000	3.706897	3.117647	3.846154	3.4375
completeness	3.583333	3.655172	3.000000	3.000000	2.7500
overall_quality	3.916667	3.793103	3.235294	3.615385	3.0625
professional_referral	3.166667	4.258621	4.235294	4.307692	2.8750
safety	4.750000	4.586207	4.352941	4.615385	4.1875

Distribution of Scores for General & Specific





While score distributions vary across models, human ratings are generally more consistent.

Conclusions & Future Work

Conclusion

- Different models show different scoring patterns across the five evaluation criteria.
- Human evaluation remained important for validating the AI generated life insurance advice.
- GPT-5.4-mini achieved the highest overall performance, but it also used more than four times as many reasoning tokens and still returned empty responses.
- Higher reasoning cost did not necessarily lead to better or more reliable results.

Future Work

- Expand the benchmark with more diverse life insurance questions and additional insurance providers.
- Evaluate newer LLMs and explore methods to improve agreement between automated and human evaluation.

References

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